

Exact Answers for Final Project Quiz 4

Modeling Profitability Instead of Default

Requires Material in Module 5 – Linear Regression

4.0 – Introduction

Your own model and the classification based on Eggertopia scores are both binary classification models – they forecast “default” or “No default.” Your bank has been reviewing individual profit and loss estimates for each customer over the three-year period and has made an interesting discovery: some defaulting customers carried so much debt for so long, and paid so much interest on it, that they lost relatively little money for the bank. Some defaulters are nevertheless net profitable for the bank! At the opposite extreme, customers who always paid off their cards in full and paid relatively little interest were not very profitable – the bank barely broke even on some of them, or even lost money.

Your boss is interested in the idea that it might be possible, instead of modeling default/no default, to model profits and losses as continuous values, using a multivariate linear regression on the same six input variables gathered on the 600 credit card recipients.

The bank could then forecast customer profitability for each customer using a scale score.

4.1 Linear Regression

If you have not done so already, standardize all six input variables from the training set. Then separately, standardize all six input variables for the test set. Standardize the profitability variable also.

Use the Excel “Linest” function to calculate the coefficients (the “betas”) that result in the best-fit line against individual profitability.

Note that the simple linear combination of six variables is probably not actually the optimum possible linear model based on these data. However, in the interests of time and reducing complexity, use only a simple linear combination of the six input variables in Part 4 of the Project.

Q: What are your values for each “beta” on the training set?*

Age	.01
Years at current employer,	.19
Years at current address,	-.07

Income over the past year, .64
Current credit card debt, -.06
Current automobile debt. 0

Do Not include the Eggertopia Scores as an input variable

Q: What is the root mean square residual error of your model on the standardized training set?*

.59

Q: What is the Correlation, R, of your model on the training set?*

.81

Q: Using the parametric formula [parametric because it assumes inputs and errors are Gaussian and continuous PDF is appropriate] $I(X:Y) = \log(1/(1-R^2))$. Assuming results on training set were sustainable (a very big assumption) what is the information gain of your linear regression model on the training set, in average bits per event?*

1.54 bits per event

Q: For the following set of inputs for 5 different applicants for credit cards, calculate the expected profit (point estimate), and the profit range that would cover 90% of outcomes (if model performance is assumed stable on future data).*

SEE ATTACHED SPREADSHEET – “profitability forecasting” with standardized data
Assume the mean profitability is \$2,000 and the standard deviation of profitability is \$5100.

Regression estimates - standardized

-.62

-.77

-.68

-.69

-.92

-\$1,161

-\$1,942

-\$1,443

-\$1,495

-\$2,668

Plus or minus

$1.67 \times .59 \times \$5100 =$

+ \$4925

- \$4925

4.2 - Testing Your Original Linear Regression Model for Over-fitting

Write an Excel function that calculates the point-estimate for each $y(i)$, applying the above *linest* “betas” from the training set to each $x(i)$ value for the six inputs.

In a separate column, you will calculate the residual between your point-estimate and the actual value for y .

The reason for this procedure is that you need to test your point estimate of y *using training set betas on the test set*. You *cannot* use Excel “*linest*” on the test set because it will simply recalculate your betas.

Q: What is the root mean square residual error of your model (with betas optimized on the *training set*), when you run it on the *test set*?*

.74

Q: What is the Correlation, R , of your model on the *test set*?*

.68

Hint: You *cannot* use Excel’s correlation function here. You may instead use the formula that for standardized data, $R^2 = 1 - (\text{the mean square residual})$.

If your model performs worse on the test set (which I expect it will) this is because the multivariate linear regression over-fits noise in the original Training Set. However, for now we will not try to correct or improve the original “*linest*” model.

Using the same parametric assumptions as above, what is the information gain of the linear regression model on the training set, in average bits per event?*

.88 bits per event

Q: Based on your new root mean square error, what is the *new* 90% confidence interval for your profitability estimates for the five applicants above?*

$$1.67 \times .74 \times \$5100 =$$

+\$6168

-\$6168